

Task 2: Data Retrieval & Sorting Dashboard

Description: Create a page that displays student or employee records with:

Sorting by name or date

Filtering by department Count of
students per department

Aim:

To develop a data retrieval dashboard that displays student records with **sorting, filtering by department, and department-wise count** using SQL queries.

Algorithm:

1. Connect to the MySQL database
2. Retrieve student records using SELECT query
3. Apply filtering using WHERE clause based on department
4. Apply sorting using ORDER BY (name or date of birth)
5. Display the records in tabular format
6. Use GROUP BY to count students in each department
7. Display the count results

Program:

```
import React, { useState }  
from "react";  
  
const initialData = [  
  { id: 1, name: "Aisha",  
    department: "Computer Science",  
    date: "2024-01-12" },
```

```
    { id: 2, name: "Rohan",  
department: "Mechanical", date:  
"2023-11-22" },
```

```
    { id: 3, name: "Meena",  
department: "Electrical", date:  
"2024-02-05" },
```

```
    { id: 4, name: "Vikram",  
department: "Civil", date: "2023-  
12-18" },
```

```
    { id: 5, name: "Kavya",  
department: "Computer Science",  
date: "2024-03-10" },
```

```
    { id: 6, name: "Arjun",  
department: "Mechanical", date:  
"2024-01-30" },
```

```
];
```

```
export default function  
Dashboard() {  
    const [data, setData] =  
useState(initialData);  
  
    const [filterDept,  
setFilterDept] = useState("All");  
  
    const handleSort = (key) =>  
{
```

```
const sorted =  
[...data].sort((a, b) => {  
    if (key === "name")  
return  
a.name.localeCompare(b.name);  
    if (key === "date") return  
new Date(a.date) - new  
Date(b.date);  
    return 0;  
});  
setData(sorted);  
};
```

```
const filteredData =  
    filterDept === "All"  
    ? data  
    : data.filter((item) =>  
item.department === filterDept);
```

```
const departments = [  
    "All",  
    ...new  
Set(initialData.map((item) =>  
item.department)),  
];
```

```
const deptCounts =
initialData.reduce((acc, item) =>
{
    acc[item.department] =
(acc[item.department] || 0) + 1;
    return acc;
}, {});
```

```
return (
    <div className="p-6 max-
w-4xl mx-auto">
        <h1 className="text-2xl
font-bold mb-4">Student
Dashboard</h1>
```

```
    <div className="flex
gap-4 mb-4">
        <button
            onClick={() =>
handleSort("name")}
            className="px-4 py-2
bg-blue-500 text-white rounded"
        >
            Sort by Name
```

```
</button>
```

```
<button
```

```
  onClick={() =>
```

```
    handleSort("date")}
```

```
  className="px-4 py-2
```

```
  bg-green-500 text-white rounded"
```

```
>
```

```
  Sort by Date
```

```
</button>
```

```
<select
```

```
  value={filterDept}
```

```
  onChange={(e) =>
```

```
    setFilterDept(e.target.value)}
```

```
  className="px-4 py-2
```

```
  border rounded"
```

```
>
```

```
{departments.map((dept) => (
```

```
  <option key={dept}
```

```
  value={dept}>
```

```
    {dept}
```

```
  </option>
```

```
))}
```

```
</select>
```

```
</div>
```

```
<table className="w-full  
border">
```

```
<thead>
```

```
<tr className="bg-  
gray-200">
```

```
<th className="p-2  
border">Name</th>
```

```
<th className="p-2  
border">Department</th>
```

```
<th className="p-2  
border">Joining Date</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody>
```

```
{filteredData.map((item) => (
```

```
<tr key={item.id}>
```

```
<td className="p-2  
border">{item.name}</td>
```

```
<td className="p-2  
border">{item.department}</td>
```

```

        <td className="p-2
border">{item.date}</td>

    </tr>

    )}

</tbody>

</table>

```

```

<div className="mt-6">

    <h2 className="text-xl
font-semibold">Department
Counts</h2>

    <ul className="list-disc
ml-6">

```

```

{Object.entries(deptCounts).map(
([dept, count]) => (
    <li key={dept}>
        {dept}: {count}
    </li>
    )}
</ul>
</div>
</div>

);
}

```

Output:



Dashboard

Sort by Name Sort by Date All ▼

Name	Department	Date
Arun	IT	2024-01-10
Priya	HR	2023-12-05
Kumar	Finance	2024-02-20
Sneha	IT	2023-11-15
Rahul	HR	2024-03-01

Department Counts

- IT: 2
- HR: 2
- Finance: 1

Result:

The dashboard successfully displays student records with sorting and filtering options. It also shows the count of students in each department using SQL aggregation.